

Data Processing & Transformation Log

PRODIOSLABS | Data Analyst Assignment

Objective

Analyze candidate behavior during an online examination using event-level telemetry logs and develop insights through SQL analysis and Power BI visualization.

Dataset Overview

Database: exam_event_logs

Table: candidate_log

Approximate Candidates: 88,000+

Event Records: 8.2M+

Examination Window: 09:00 AM – 07:15 PM

Data Validation

- Total row count verification
- Distinct candidate count verification
- Activity distribution analysis
- Language distribution verification
- Question section validation
- Question response validation
- Missing value inspection
- Composite key validation (log_id + candidate_id)

Data Quality Observations

- Approximately 90.73% of records were Auto Save events.
- Auto Save events represent system-generated logs rather than direct candidate actions.
- No answer keys were provided, preventing correctness analysis.

Data Transformation Steps

Activity Analysis: Grouped event records by activity type to identify dominant user and system actions.

Candidate Segmentation: Candidates were segmented using event frequency. Since Auto Save accounts for the majority of records, segments should be interpreted as event-volume categories rather than direct measures of engagement.

Section Analysis: Aggregated interaction counts by question section to compare section-wise interaction volumes.

Language Analysis: Grouped records by question language and section to analyze language distribution across sections.

Response Analysis: Analyzed recorded response states (A, B, C, D). Results were not interpreted as preference or correctness due to Auto Save dominance and the absence of an answer key.

Time-Based Analysis: Aggregated events by hour to identify event-volume trends during the examination window.

SQL Queries Used

Activity Analysis
Review Behaviour Analysis
Candidate Segmentation
Section Analysis
Section-wise Language Analysis
Response Distribution Analysis
Hourly Event Analysis
Session Duration Analysis
Composite Key Validation

Power BI Visuals

KPI Cards
Clustered Bar Charts
Stacked Column Charts
Line Charts
Slicers

Assumptions

- Auto Save events were treated as system-generated interactions.
- Response states do not indicate correctness.
- Event volume should not automatically be interpreted as candidate engagement.